Architecture Design for Turing Test and CAPTCHA Systems

By: Abhinav Reddy Pakanati,SE24UCSE129

The Turing Test, evaluates a machine's ability to show intelligent behavior indistinguishable from that of a human through conversation. In a practical implementation, the system follows a client-server architecture. An appropriate architecture for implementing the Turing Test includes a user interface where a human judge communicates anonymously with two participants, one human agent and one AI agent, through a conversation controller that routes messages without revealing identities. The AI agent uses a natural language processing engine to generate responses, while the system logs interactions and includes an evaluation module where the judge determines which participant is human.

In contrast, CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) is designed to distinguish humans from bots automatically. Its architecture consists of a web application interface connected to a CAPTCHA generator that creates challenges such as distorted texts and image recognition tasks. These challenges are stored securely with hashed answers and expiry timestamps in a challenge database, and a verification engine validates user responses before granting or denying access.While the Turing Test focuses on measuring artificial intelligence through human judgment, CAPTCHA systems emphasize automated security mechanisms to prevent unauthorized automated access.